# COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit – 1 : Session – 1 : SLO – 2 | Types of computers
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## Multiple Choice Questions

• 1. c — Supercomputer
• 2. c — High throughput and multi-user support
• 3. b — Built into another device for a specific task
• 4. c — Personal laptop
• 5. c — Speed and reliability

## Two-Mark Questions

• 1. A minicomputer is a mid-range, multi-user computer smaller than a mainframe but more powerful than a microcomputer; it supports several terminals at once. Real-time application: process control in a factory / college server.
• 2. General-purpose vs special-purpose: (i) General-purpose computers run many different applications (PCs); special-purpose computers are designed for one dedicated task. (ii) General-purpose machines are flexible but less optimized; special-purpose machines are faster/cheaper for their single task but not reprogrammable for other work.

## Scenario-Based Long Answer

• Recommended: a Mainframe-class server (a high-end minicomputer is the budget alternative).
• Justification: The university needs simultaneous access by many users (student records, exam processing, library). Mainframes are built exactly for this: very high throughput, strong multi-user support, large memory and I/O capacity, high reliability/availability, and transaction processing at scale. A minicomputer could serve a smaller load, but for a mid-sized university with concurrent access across departments, a mainframe's multi-user capability and reliability make it the most suitable choice.

## Match the Following

• Supercomputer — c) IMD weather forecasting
• Mainframe — large multi-user transaction servers (e.g., bank/airline reservation systems)
• Minicomputer — b) College server
• (Smartwatches — a) — are examples of embedded/microcomputer systems.)

## Fill in the Blanks

• 1. Supercomputer 2. Special-purpose 3. Embedded 4. microcomputers 5. Ubiquitous (pervasive)

## Scenario-Based Small Activity

• 5 devices with embedded computers: smartphone, washing machine, microwave oven, smart TV, car (ECU).
• Example — washing machine: its embedded controller runs a fixed control program; it reads sensors (water level, load, temperature), times each wash/rinse/spin stage, and drives the motor and valves automatically. It is special-purpose: it does only washing control, reliably and cheaply.

## True or False

• 1. FALSE — Microcomputers serve one/few users. 2. FALSE — Special-purpose are less flexible.
• 3. TRUE 4. FALSE — Minicomputers serve departments/organizations, not national defense scale.
• 5. TRUE